

Business process management and digital transition. The case study of an Italian Public University

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Abstract

Purpose – The purpose of this study is to highlight how the digitalization of a public university through a structured Business Process Management (BPM) approach allows for a significant performance improvement, even in a bureaucratized context not inclined to process thinking.

Design/methodology/approach – The used research methodology centers on a single case study conducted at an Italian public university. The selected process has been examined and redesigned within the BPM lifecycle framework. The AS-IS and TO-BE state, i.e. before and after the organizational change, have been modeled according to BPMN2.0 notation and evaluated through quantitative and qualitative techniques.

Findings – The authors first carried out a literature review to identify pertinent performance indicators suitable for assessing a BPM project within a public organization. Secondly, applying the BPM framework to the case study enabled significant improvements in the quality of the process. Third, the authors analyzed the impact on people and the organization and how to soothe the transition to a digitalized process.

Originality/value – The study's findings can contribute to the existing body of knowledge on BPM as a digitalization approach in the public sector. The case study is the first of its kind in the higher education context. Its value also resides in highlighting the efficacy of using BPM as a guiding tool for making organizational and technical decisions during the implementation phase within the specific context of this paper.

Keywords Digital innovation, eGovernment, Higher education, Information systems integration, Performance optimization, Digitalization

Paper type Case study

1. Introduction

In an ever-changing competitive landscape, organizations must continuously innovate (Santarsiero *et al.*, 2022) and digitalization remains an important innovation driver, capable of providing businesses with significant competitive advantages (Ammirato *et al.*, 2019a). Digitalization refers to the integration of digital technologies into everyday processes, leading to fundamental changes in how organizations operate and deliver value to customers. Digitalization is becoming essential to obtain better performance (Hellsten and Paunu, 2020). It leads to several benefits:

- faster decision-making;
- a better quality of information;
- improved user experience;
- efficiency;



- coordination among processes;
- transparency; and
- easier accessibility of information (Agostino and Costantini, 2022; Costa Melo *et al.*, 2023; Hellsten and Paunu, 2020; Kuusisto, 2017; Terlizzi, 2021).

At the macro level, digitalization initiatives have been carried out worldwide, such as the Digital Europe Programme [1], Smart Africa (2020) and Smart Nation Initiative [2]. These initiatives aim to reach digital transformation within and across both public and private sectors, boosting national economies.

At a micro level, organizations that will exploit and benefit from digitalization need a structured approach to manage change in their business models (Bresciani *et al.*, 2021). Toward this aim, different authors recognize the potential of Business Process Management (BPM) in digitalization (Antonucci *et al.*, 2021; Kirchmer, 2017; Stjepić *et al.*, 2020).

BPM is a managerial approach to making an organization's processes more efficient and dynamic to meet the changing needs of stakeholders by leveraging digital technology adoption (Dumas *et al.*, 2018). It consists of a range of methods, techniques and tools that identify, analyze, redesign, optimize, execute and monitor business processes and their interactions (Macedo De Moraes *et al.*, 2014). BPM approaches boost the benefits of digitalization in terms of improved customer satisfaction, efficiency, timeliness, service quality, compliance and transparency (Christiansson and Rentzhog, 2020; Do Nascimento *et al.*, 2020; Hammer, 2015; vom Brocke *et al.*, 2014). Historically, BPM has found wide application in the private sector, which is farther ahead in successful BPM implementation as compared to the public sector (Kissa *et al.*, 2023).

However, BPM is beneficial for public companies as well in that it enhances accountability, quality and efficiency (Battilani *et al.*, 2022; Terlizzi, 2021). It is crucial for addressing citizens' evolving needs and increasing responsiveness (Terlizzi, 2021) given that lately, the public sector has become more customer-oriented (Ongaro, 2004; Indihar Štemberger and Jaklič, 2007). A BPM approach can, in principle, serve as an effective solution for digitalizing the public sector. Nevertheless, the application of BPM in this field is relatively new (Van Looy and Van den Bergh, 2018) and mandates further insights since few interventions are testified by the scientific literature (Kissa *et al.*, 2023).

This paper aims to demonstrate the importance of a structured BPM approach in process digitalization as a means to achieve sensible improvements for organizations in terms of operational efficiency. A case study is therefore presented regarding the improvement of the business trip request and approval (BTRA) process: an internal process within an Italian Public University, following the methodological approach of the *BPM life cycle* as proposed by Dumas *et al.* (2018).

By introducing the case study, we aim to demonstrate to what extent the BPM approach positively affects digitalized process performance. The value of the contribution to the existing body of knowledge lies in the application of a well-defined approach to a specific process of the public sector, using well-understood and effective process improvement techniques. This study covers all phases of the lifecycle up to the implementation of the redesigned process, filling the gap of a detailed exploration in BPM case studies, particularly its impact on organizations and people. In this paper, we provide insights into these aspects and describe how we managed the digitalization aspects. Additionally, the actual implementation resulted in significant improvements in operational efficiency, consistent with previous studies.

The paper is organized as follows : Section 1 delineates the motivation of the study. In Section 2, an overview of the theoretical background and the policy background is presented.

Section 3 and Section 4 illustrate the methodology and empirical findings, respectively. Section 5 is a discourse on the discovered results, while Section 6 reports the concluding remarks while addressing implications, limitations and future research.

2. Theoretical background

Throughout the paper, we use the term “digitalization”. It is different from “digitization” which refers to the conversion of “analog information into digital information” and focuses on digitalizing documentation processes without affecting value-creation activities (Elia *et al.*, 2024; Mergel *et al.*, 2019; Verhoef *et al.*, 2021). Digitalization represents a step further since it involves the implementation of digital technologies “to alter existing business processes” (Berman, 2012; Mergel *et al.*, 2019; Verhoef *et al.*, 2021). Henceforth, we refer to digitalization since it affects and modifies business processes, leveraging digital technologies as an enabling factor to sensibly improve process performances.

2.1 The digitalization of the public sector

Following Adami’s (2002) definition of complexity, public sector processes are complex since they rely on extensive information. Increasing this complexity can improve their adaptability and resilience to change, to which the public sector is prone. Within this context, the importance of digitalization in bolstering management and use of information cannot be overlooked. The relevance of public sector digitalization has attracted the interest of academics and policymakers, while a more efficient and productive (Umbach and Tkalec, 2022) public sector fosters a nation’s economic development, competitiveness and trustworthiness (Terlizzi, 2021). The digitalization of the public sector has been explored globally, with varying degrees of success. Examples include Australia’s Digital Government Strategy [3], the UK’s Government Digital Service Strategy (GOV.UK, 2022), Estonia’s e-Estonia program (e-Estonia, 2021), the South Korean e-Government Master Plan (OECD, 2021), the US Digital Government strategy [4] and “Piano Triennale per l’Informatica” in Italy (AGID, 2023). These initiatives highlight the role of structured approaches to digitalization in enhancing efficiency, transparency and service quality in the public sector and share the same operational goals:

- ensuring online availability of main public services;
- implementing easy-to-navigate government portals;
- carrying out cyber security plans;
- creating unique digital identities for citizens;
- employing digital signatures; and
- digitalizing education and health infrastructures, and so on.

Many countries worldwide have already undergone digital transformation endeavors, seeing successes and failures. Successful initiatives have been testified (Australian Government, 2024; GOV.UK, 2023; Himma-Kadakas and Kõuts-Klemm, 2023; Digital Government, 2021; Scupola, 2019). A particular successful government initiative is being conducted in the UK, where the government 2022–2025 roadmap aims to improve government operations and the citizens’ digital experience in accessing public services, leveraging cutting-edge technology to keep up to date, implementing a progress tracking system and improving the quality of data (GOV.UK, 2023).

The government manages to reach some of the operational goals listed above, such as the One Login, enhanced efficiency, usability and safety of public services, along with citizens’

digital upskilling promotion. The Australian Government is another successful story. It has an Implementation Plan, that outlines the actions to digitalize the government and includes metrics to track the progress of the “Data and Digital Government Strategy” (Australian Government, 2024). It aims to enhance public services by enhancing legacy ICT systems, the Digital ID system and facing cyber-attacks (Australian Government, 2024).

However, these successful experiences lag in some sectors, and the level of digitalization is not uniformly distributed within countries. For example, in Australia, more remote regions and certain demographics tend to lag in digital advancements (Australian Government, 2024). There are also instances of digitalization failure, particularly in developing countries, where various challenges impede progress (Syed *et al.*, 2023; Aminah and Saksono, 2021).

Italy is a case of academic interest in this regard. It has been underscored by Eurostat (2023) statistics and the Report on the State of the Digital Decade (European Commission, 2023), which provides a holistic evaluation of the digital progress of the European Union members. Italy stands out in digitalization of the public sector, especially when put in contrast to the other EU countries. Although there has been a remarkable improvement in recent years in Italy, mainly due to the sprint caused by the COVID-19 pandemic (Terlizzi, 2021), the performance in public sector digital services is worse than the European average. Hence, Italy should carry on improving public digital services more rapidly.

Besides the ever-evolving digital environment, these disparities and setbacks explain the ongoing commitment to digitalization and underscore its critical importance for state governance. This highlights the importance of dealing with the public sector digitalization, its societal implications, and arising benefits.

2.2 Business process management for digitalizing the public sector: insight from a literature review

In literature, various case studies about public sector digitalization follow different methodological approaches, such as Lean (Goretti *et al.*, 2023), Lean Six Sigma (Meza and Jeong, 2013) and Enterprise Architecture approach (Ajer and Olsen, 2018). However, we chose the BPM approach, since it is more operational and focused on managing business processes, thus best-fitting the characteristics of the proposed BTRA case study, such as the need for flexibility and compliance (Kittel and Sackmann, 2011).

The scientific literature recognizes the BPM’s role in improving the public sector’s performance in terms of customer satisfaction, efficiency, timeliness, service quality, compliance and transparency (Christiansson and Rentzhog, 2020; Do Nascimento *et al.*, 2020; Hammer, 2015; Kirchmer, 2017; vom Brocke *et al.*, 2014). As stated by Syed *et al.* (2018) BPM is “an enabler for high performance and increased customer centricity”. These are attributes of paramount importance in the public sector (Cordella and Bonina, 2012; Ufua *et al.*, 2021), which is a fertile ground for BPM’s implementation because its rigid structure and organizational culture are resistant to radical changes (Indihar Štemberger and Jaklič, 2007).

To explore the use of BPM to digitalize the public sector, with particular attention to the higher education field, we conducted a systematic literature review. To the best of the authors’ knowledge, no previous studies present in the literature offer a precise overview of the topic under investigation and thus provide insights on how to carry out the case study and best set up the technical and organizational variables. Therefore, we performed a systematic literature review as a first attempt to highlight successful case studies, applying the BPM approach to the public sector’s digitalization and performance improvement. More precisely, we aimed to identify the KPIs (Key Performance Indicators) that best suit a BPM intervention in the public sector together with technical and organizational details on how

digitalization can be carried out. According to the indicators in [Denyer and Tranfield \(2009\)](#) and [Ammirato et al. \(2023\)](#), the research protocol we adopted consisted of three main steps:

- (1) paper location and selection;
- (2) paper analysis; and
- (3) results presentation.

2.2.1 Paper location and selection. We performed our search on Scopus scientific database, due to its acknowledged relevance in the managerial field of study ([Bhimani et al., 2019](#)). We queried the database, searching within article, title, abstract and keywords, as follows: (“Business process re-engineering” OR “BPR” OR “Business process management” OR “BPM” OR “Business process change” OR (“Digitalization” AND “BPM”) OR (“Digitalization” AND “Business Process Management”)) AND (“Public” OR “Higher education” OR “Government”) AND “Case study”. We conducted the research at the end of November 2023, finding 247 documents. Papers not written in English and conference papers were excluded from the sample, and we limited the results to Business, Management and Accounting and Engineering subject areas. Then, we retrieved 87 papers which we analyzed by their title and abstracts. In the end, 11 papers constituted the final sample for the deeper study as they adhere to the aim of the literature review (see [Table 1](#)). In [Figure 1](#), the selection process has been depicted.

2.2.2 Paper analysis. We carefully read and analyzed the 11 papers included in the final sample to collect information about BPM case studies in the literature and investigated their potentiality and implementation. The analysis shows a shift during the years from radical approaches, typical of the Business Process Reengineering, to more transformational approaches proper of the Continuous Improvement approach BPM. Even when the authors refer to a BPR ([Hesson, 2007](#)), they implement a less radical approach, like BPM. This is because the public sector generally rejects radical change ([Indihar Štemberger and Jaklič, 2007](#)) and tends to freeze its processes if there are not too many critical points ([Hesson, 2007](#)).

Not all the sampled case studies focus primarily on process digitalization details, which is the focus of this paper. However, in those papers that follow a BPM approach to digitalize processes, its effectiveness is clear. Many papers emphasize the redesign of the current processes without a thorough investigation of the implementation phase, thus avoiding details on how to operationalize digitalization and the necessary change management. However, understanding how to foster change acceptance by stakeholders is crucial, as it is one of the critical success factors ([Trkman, 2010](#)).

2.2.3 Results presentation. The analysis led to identifying the most appropriate performance indicators for evaluating public organizations (see [Table 1](#)).

As reported in [Table 1](#), although performance factors are sensitive to the context’s characteristics, e.g. administrative traditions ([Umbach and Tkalec, 2022](#)), the KPIs used to assess the process refer mainly to time-related metrics ([Battilani et al., 2022](#); [Bisogno et al., 2016](#); [Hesson et al., 2007](#); [Kissa et al., 2023](#); [Pereira et al., 2020](#); [Renna and Izzo, 2018](#); [Renna et al., 2020](#); [Rinaldi et al., 2015](#)). Even in papers that do not explicitly refer to the use of KPIs, time is the factor taken most into consideration to evaluate the effectiveness of a process ([Do Nascimento et al., 2020](#); [Hesson, 2007](#)). Other relevant performance dimensions are cost, quality and flexibility. In the only paper that refers to cost is ([Kissa et al., 2023](#)), the authors measure it, depending upon the duration of each activity, confirming the importance of time for performance evaluation. Quality and flexibility are, instead, not evaluated since they are hard to compute but influenced reflexively by other performance dimensions.

Table 1. Analyzed papers

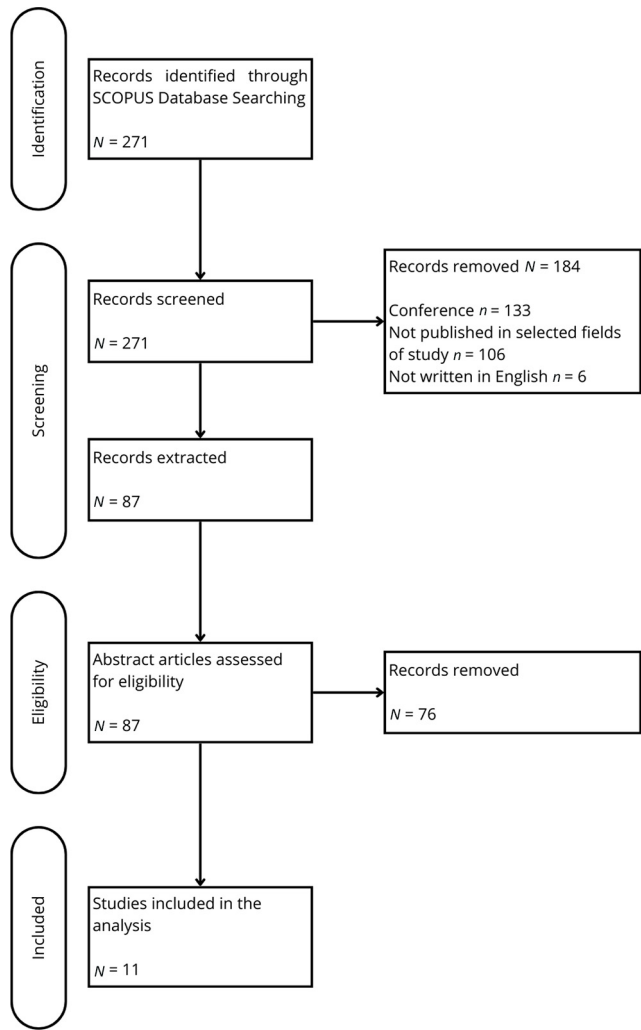
Authors	Title	Year	Source	KPI
M. Hesson	Business process reengineering in UAE public sector a naturalization and residency case study	2007	<i>Business Process Management Journal</i>	Processing time
M. Hesson, H. Al-Ameed, M. Samaka	Business process reengineering in UAE public sector a town planning case study	2007	<i>Business Process Management Journal</i>	
M. Indihar stemberger, J. Jaklic	Towards E-government by business process change – A methodology for public sector	2007	<i>International Journal of Information Management</i>	
M. Rinaldi, R. Montanani, E. Bottani	Improving the efficiency of public administrations through business process reengineering and simulation: a case study	2015	<i>Business Process Management Journal</i>	
S. Bisogno, A. Calabrese, M. Gastaldi, N. Levialdi Ghiron	Combining modeling and simulation approaches	2016	<i>Business Process Management Journal</i>	
P. Renna, C. Izzo	Using business process management simulation to support continuous improvements in higher education management system	2018	<i>International Journal of Management in Education</i>	• Maximum number of users in queue • Total average waiting time • Average employee saturation • Completion rate of process • Throughput time • Rate of resource utilisation • Resource service level • Average utilization of the main resources • Throughput of the students' instances • Average time to complete the instances
R. Pereira, L.V. Lapão, I.S. Bianchi, D. Amaral	Improving emergency department through business process redesign: an empirical study	2020	<i>Australasian Journal of Information Systems</i>	• Average process time durations • Maximum process time durations • Instances started • Instances completed

(continued)

Table 1. Continued

Authors	Title	Year	Source	KPI
A.R. do nascimento, R. Baldam, L. Costa, T. Coelho	Applications of business governance and the unified BPM cycle in public credit recovery activities	2020	<i>Business Process Management Journal</i>	
P. Renna, C. Izzo, T. Romaniello	The business process management systems to support continuous improvements	2020	<i>Handbook of Research on Quality Assurance and Value Management in Higher Education</i>	• Total completion time • Average time for unit • Maximum time for unit • Total number of instances completed over a period • Average resources' utilisation • Resource saturation • Crossing times of the process
C. Battilani, G. Galli, S. Arecco, R. Revetria, L. Damiani	Business process re-engineering in public administration the case study of Western Ligurian Sea Port Authority	2022	<i>Sustainable futures</i>	
B. Kissa, E. Gounopoulos, M. Kamariotou, F. Kitsios	Business process management analysis with cost information in public organizations: a Case Study at an Academic Library	2023	<i>Modelling</i>	• Processing time • Average cost

Source: Authors' own work



Source: Authors' own work

Figure 1. Research workflow

Although the theme of digitalization is not extensively explored in these papers, overall, BPM is acknowledged as a digitalization enabler (Battilani *et al.*, 2022; Do Nascimento *et al.*, 2020; Hesson *et al.*, 2007; Pereira *et al.*, 2020) due to its suitability for process digitalization (Ahmad and Van Looy, 2020). BPM defines a structured framework to facilitate IT innovation projects whose “main goal is to amalgamate people and automated systems within structured process workflows to reach the best performance values” (Ammirato *et al.*, 2019a).

However, even if the combination of digital innovation and BPM can be beneficial, studies on digital innovation in the administrative and management fields, particularly within

the context of higher education (Duarte and Martins, 2014), are scarce, corroborating the crucial role of digitalization in this context (Kleider *et al.*, 2021). Among the papers chosen, only a few (Kissa *et al.*, 2023; Renna and Izzo, 2018; Renna *et al.*, 2020) refer to this area but do not focus on process digitalization. Unfortunately, the sampled papers do not present technical and organizational details on how digitalization can be carried out that could be used as best practices in the implementation phase of the case study.

Also, if we focus specifically on the Italian case, the literature shows a lack of contributions related to BPM case studies to digitalize the Italian public sector (Bisogno *et al.*, 2016; Renna and Izzo, 2018; Renna *et al.*, 2020; Rinaldi *et al.*, 2015). For instance, in (Rinaldi *et al.*, 2015) the improvement does not involve the digitalization of the process, since it proposes changes in the employees' work shifts. Among the papers in Table 1, only one (Battilani *et al.*, 2022), the most recent in the Italian context, deals with the public sector's digitalization of seaport processes. However, this case is quite different from the education context that our paper aims to examine: it implements the BPR approach, which is less structured than the BPM lifecycle.

3. Methodology

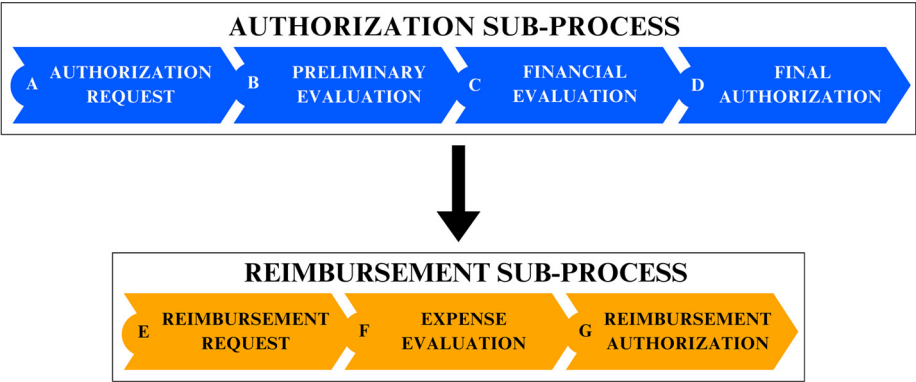
3.1 Research design

The research adopts a case study methodology to explore the domain and implement a structured approach. As stated by Gerring (2004), the case study is "an intensive study of a single unit to understand a larger class of (similar) units". It has descriptive, theoretical testing and theories' generation from practice capabilities (Benbasat *et al.*, 1987; Eisenhardt, 1989). Using archival data and interviews as evidence, the case study is a suitable approach for investigating a contemporary phenomenon (Yin, 2003). This methodology transposes its application in a real context to deeply understand its framework and acquire a further understanding of the background, overcoming the scarcity of available information. It leads to finding concrete improvement opportunities. Even if the case study's findings are not generalizable, it led to the generation of new hypotheses (Sellitto, 2018). The approach itself led to a better investigation of the academic complex context, in which there are several mandatory laws and regulations to follow. To delve deeper into this relatively underexplored domain, we chose the methodology of a single-case study for investigation.

3.2 Case selection

The case study relates to the digitalization of the BTRA process within an Italian university. Italy's digital position among European nations justifies this regional setting. Despite the continuous government efforts to support the public sector's digitalization (AGID, 2023; MEF, 2021), most public services remain ineffective and not digitally supported (European Commission, 2023).

As stated, this work refers to a field of application that has not been covered in other papers. The BTRA process is a case of scientific and practical interest as it is shared across multiple organizations and is scarcely digitalized. Typically, it is made of two sub-processes: the authorization and the reimbursement (see Figure 2). The authorization sub-process encompasses the authorization request phase (A), the preliminary evaluation phase, in which a manager evaluates the institutional opportunity and convenience of the trip (B), the financial assessment phase, to verify fund availability to cover trip expenses (C) and the final authorization (D). The reimbursement sub-process consists of the reimbursement request phase (E), the evaluation of the expenses' refundability (F) and the authorization phase to issue the reimbursement (G). Specifically, the university context is justified by the data availability and accessibility, easing a deepened analysis (Yin, 2003).



Source: Authors' own work

Figure 2. BTRA process

3.3 Data collection and analysis

The study implements the BPM lifecycle, introduced by [Dumas et al. \(2018\)](#), into a real case study to improve processes and validate its feasibility. It is composed of the following phases:

- Process identification: includes the definition of the process architecture and the process selection.
- Process discovery: concerns team definition, process detection and quality control of the modeled process. The output is the AS-IS model, namely the current process workflow.
- Process analysis: aims at identifying the weaknesses of the process and their impact, both qualitatively and quantitatively.
- Process redesign: involves the improvement of the business process to reach business objectives. This phase produces the TO-BE model, namely the improved process.
- Process implementation: concerns the implementation of the developed model. It also involves the change management process.
- Process monitoring: involves the continuous monitoring of the implemented process. The ensuing data are analyzed to get further improvements.

It is an iterative cycle, from monitoring to discovery phases, which should be periodically repeated within a company to maintain high standards.

4. Findings

This section delineates the outcomes arising from the investigation conducted on the case study.

4.1 Identification phase

Among the processes occurring in a university as stated by ([Kahloun and Ayachi-Ghannouchi, 2020](#)), this case study concentrates on improving a support process, expressly

an administrative one. This class of processes is known for their transformative potential, given their impact on the organizational dynamics and relative ease of redesign (Naplava *et al.*, 2014). The chosen process is the BTRA process.

4.2 Discovery phase

During this phase, we collected data through online documentation analysis and structured interviews with domain experts. The process, encompassing authorization and reimbursement sub-processes, involves multiple actors. The applicant submits the business trip application. The head of the department, general manager, vice-rector or dean has top clearance for the final authorization phase, based on the applicant type. Depending upon the applicant's role and the fund's typology, the scientific director, the PhD coordinator, the fund's responsible and the head of the fund affiliation unit execute the previous authorization phases. The latter also conducts the reimbursement authorization phase, preceded by the department secretary.

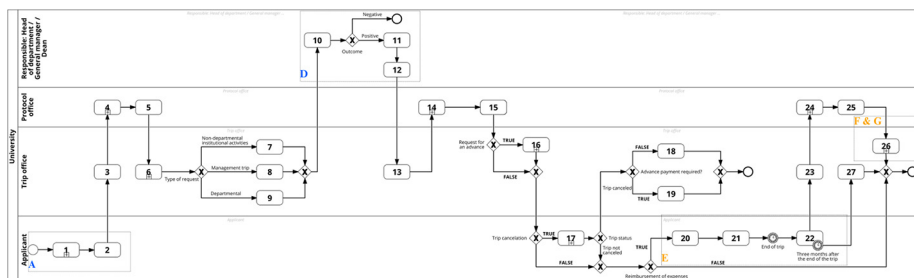
Other actors involved in the BTRA process are the trip office that manages support activities and evaluates expenses' refundability, and the protocol office that registers applications and authorizations for document management. The entire process involves three information systems (ISs), which respectively manage budgeting aspects of the trips, provide treasury services and manage electronic document workflow.

This investigation identified the elements necessary to model the process through a BPMN diagram, known for its readability (Bisogno *et al.*, 2016; Kahloun and Ayachi-Ghannouchi, 2020).

Figure 3 illustrates the overall process, decomposed into further sub-processes, modeled, using BPMN2.0 language (www.bpmn.org). Figure 4 illustrates the main sub-process, with other sub-processes omitted to avoid text overload. The process aligns with the main phases outlined in Section 3.2, as evidenced by the framed activities associated with each phase. Activities and collapsed sub-processes are identified by acronyms for readability, with their name linked to acronyms in Appendix 1.

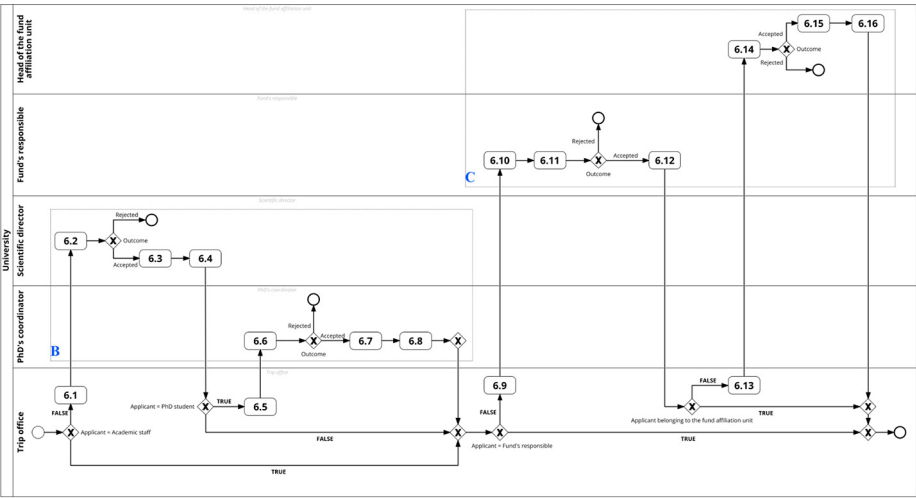
4.3 Analysis phase

The analysis phase is pivotal for identifying weaknesses in the current process, devising interventions and evaluating changes' impact. The chosen tools are value-added and flow analyses (Ammirato *et al.*, 2019b; Barmayoun and Mocan, 2020; Dumas *et al.*, 2018; Kissa *et al.*, 2023). Value-added analysis is a qualitative technique to identify redundant activities



Source: Authors' own work

Figure 3. AS-IS Model's main process



Source: Authors' own work
Figure 4. AS-IS model – finalize first authorization steps sub-process

for elimination or digitalization. Flow analysis is a quantitative technique to analyze a process referring to a performance indicator. In this study, we choose the cycle time, which is the average time an instance of a process takes from its beginning to its ending. [Appendix 1](#) illustrates activity durations and first analysis results, categorizing activities as non-value adding (NVA), business value adding or value adding (VA) based on their value for the process' customer.

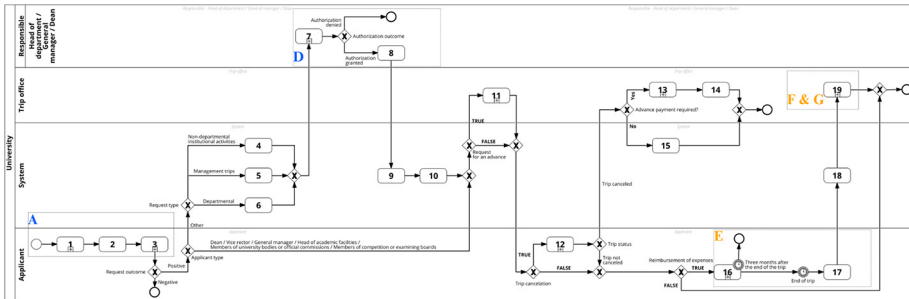
Domain experts provided cycle time estimates. Through interviews and analysis, the main weaknesses were highlighted: increased throughput time due to manual application transfers, non-compliance with laws such as d.lgs. 82 / 2005, and later changes, regarding the need for public organizations to provide online services based on effectiveness, cost-effectiveness and usefulness criteria.

4.4 Redesign phase

The redesign phase aims to innovate the AS-IS model, ensuring strict adherence to new regulations and delivering an elevated level of service to stakeholders. From this perspective, a digitalization strategy is implemented to address the weaknesses of paper-based processes observed during the previous phase.

To reach these objectives, NVA activities are automated or eliminated. Furthermore, the BTRA process is unified and digitalized across university departments. However, since the implementation of a platform for all departments is an extensive intervention, the TO-BE model was preceded by its implementation only in a single department. Although this gradual approach took longer, it has significantly mitigated the probability of failure of the intervention, leading to the preparation of a suitable contest for change adoption.

[Figure 5](#) illustrates the main process within the TO-BE model, which involves a system to manage the entire life cycle of the process, while [Figure 6](#) illustrates its main sub-process. As for the AS-IS, we traced it back the TO-BE model to the phases of the general process as outlined in [Section 3.2](#). [Appendix 2](#) shows the acronyms of the activities, their expression in

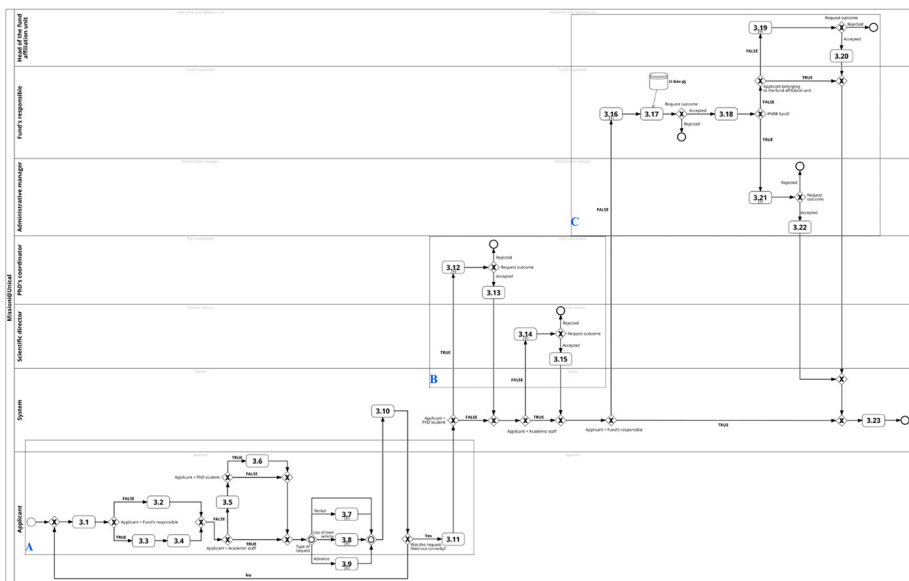


Source: Authors' own work

Figure 5. TO-BE model's main process

natural language and their duration in the redesigned process, based on estimations of domain experts that were gathered after the implementation of the redesigned process.

The consultation through periodic meetings of key users led to the user interface and functionalities implementation. Beyond the digitalization of the process, the TO-BE model extends its purview to encompass business trips financed by PNRR funds. Consequently, the model includes another actor: the administrative managers, who verify the requests before the final authorization phase. The flow analysis is reiterated to evaluate the redesigned process.



Source: Authors' own work

Figure 6. TO-BE model – request trip authorization sub-process

4.5 Implementation phase

According to the redesigned version, we use new digital technologies to optimize the process during the implementation phase. The TO-BE workflow became the basis for the new digital platform development, named Missioni@Unical. It streamlined the execution of the process, referring to a single platform to carry out the activities to enhance overall performance. To reach the goal of a complete process digitalization, Missioni@Unical integrates with the pre-existing IS involved in the process, such as the document management system (DMS), the single-sign-on (SSO), the digital signature, the personnel database and the accounting IS. The DMS manages electronic document flows (AGID, 2018a). The SSO enables users to access the system and authenticate themselves. “The digital signature allows users to exchange documents online with full legal validity” (AGID, 2018b).

The accounting system information integration allowed the platform to hold updated data about projects and available funds. Missioni@Unical implements a role-based access control system, assigning different privileges and actions to each user according to his role. This is due to integrating the organization’s IS and the authorization hierarchy. ISs integration brings several benefits, as information is shared automatically between all the IS, avoiding human intervention for manual insertion while ensuring reliable and coherent information among all systems.

4.5.1 Impact of digitalization on people and organizations. The primary factor influencing the success of a project in the public sector, to which the university under study belongs, is the resistance to change among the involved stakeholders (Jurisch *et al.*, 2013; Lašáková *et al.*, 2017), which can nullify the potential benefits of digitalization. During the redesigning, potential resistance from the public organization was already evident. Therefore, a change management process became necessary. Hence, following the implementation phase, the project team to undertake several actions to overcome the resistance:

- presentation of the platform’s functionalities to the academic personnel and back-office staff;
- presentation department-by-department for the academic personnel as part of departmental council meetings;
- training process specific to the administrative staff;
- availability of tutorials to support users in self-learning and self-solving issues and a FAQ section;
- creation of a how-to-use section within the platform, covering each process step; and
- establishment of a tiered help desk system, encompassing a functional level, where a trained staff address usage issues and a technical level involving system developers for resolving technical bugs.

From an organizational point of view, the digitalization intervention affected the organization structure at a micro rather than macro level, as it had not undergone any changes; there had been only a shift in the resources’ workload. The university under examination has 14 departments, and each department has one full-time personnel unit dedicated to the BTRA process and two personnel units handling the process in the back office. After the digitalization, the overall workload on the trip office had been reduced by almost 60% in terms of workforce, as can be deduced referring to this office workload moving from the AS-IS to the TO-BE model (see Appendix 1 and Appendix 2). This workload decrease led to the reduction of the workload of the personnel involved in the

process. Therefore, the university reallocated the personnel units to other more critical activities.

4.6 Process monitoring (performance analysis)

We compared the calculated metrics to evaluate the performance enhancement. In Table 2, 12 instances are identified, based on the most frequently activated paths. We calculated the cycle time, considering activities that belong to each path. The underlying assumption is that each authorization step has a positive outcome. Table 2 delineates the cycle time of the AS-IS and the TO-BE models highlighting the time saving. The comparative analysis of cycle time reveals an average saving of approximately 55% with a peak of 79,6%.

Notably, the best results are obtained in the authorization sub-process, as the applications exclusively comprising the authorization phases exhibit the highest savings. This achievement is due to the heightened efficiency in form completion and the automated transmission among stakeholders, thanks to the system. Furthermore, the savings are due to downtime elimination, thanks to the automation of activities.

5. Discussion

The public sector plays a pivotal role in a country's development. Consequently, it becomes imperative to continually renovate itself, even if public organizations usually fear change of the status quo. Hence, the effort needed does not concern only digitalization but also the creation of a conducive environment and the implementation of an effective change management process. Indeed, the empowerment of stakeholders is crucial to the success of these projects (Umbach and Tkalec, 2022).

The primary objective of this study was to demonstrate that BPM lifecycle implementation can guide digitalization efforts and enhance the operational efficiency of the process by reducing the throughput time. This supports the findings in previous studies such as (Hesson *et al.*, 2007; Bisogno *et al.*, 2016; Renna and Izzo, 2018; Pereira *et al.*, 2020; Renna *et al.*, 2020; Battilani *et al.*, 2022). It also promotes the change applied to meet new regulations. Simultaneously, the changes implemented brought about several indirect benefits, identified through direct and participant-observation and stakeholder' interviews as recommended (Yin, 2003).

Our study aligns with the existing literature, but it adds depth by analyzing the implementation phase while suggesting how to manage the human factor that can lead a project to failure. The study identified KPIs such as cycle time, processing time and user satisfaction to evaluate the BPM project's effectiveness. The application of BPM in digitalizing the BTRA process resulted in several organization benefits, the main one being a 55% average reduction in cycle time. Additionally, stakeholder satisfaction improved due to enhanced transparency and reduced administrative burden.

Other important operational results emerge from the digitalization:

- *The pursuit of the once-only principle narrowed to the university.* The system authenticates the user and automatically populates the requisite fields with its data. Furthermore, users can save pertinent information on the platform eliminating the need for repetitive data entry, aligning with principles pursued in previous work such as (Hesson, 2007).
- *Paper savings.* Digitalization allows the actors to carry out the process through digital instruments, thus avoiding paper waste.
- *Ability to access the system from anywhere and at any time and higher transparency.* The applicant can access the system anytime and anywhere, providing flexibility

beyond office hours. Digitalization and dematerialization of the process lead to higher transparency and stakeholder awareness. Applicants can monitor the real-time progress of their applications. It aligns with (Kuusisto, 2017; Terlizzi, 2021), who outlined the importance of transparency, online accessibility and their link to digitalization.

- *Higher quality of the service and degree of user and employee satisfaction.* It derives from the reduced service time, enhanced access to information and real-time monitoring of application progress. The applicant receives prompt e-mail notification regarding the progression or impediment at significant process junctures. Process automation not only streamlines throughput time but also removes redundant activities, ensuring enhanced quality and satisfaction levels among employees and users alike (Ongena and Ravesteyn, 2019), a benefit achieved in other papers such as (Hesson, 2007; Hesson *et al.*, 2007).

These improvements are facilitated by platform integration with the involved ISs, reducing cycle time, and enhancing overall process execution.

The improvements obtained also align with the objectives of the “Piano Triennale dell’Informatica” (AGID, 2023), such as the digitalization of processes, the adherence to the once-only principle and the enhancement of the user experience.

6. Conclusion

The research methodology of this paper centers on a single-case study approach focusing on an Italian public university’s BTRA process. The BPM lifecycle (Dumas *et al.*, 2018) was used as a structured methodological approach to digitalization encompassing process identification, discovery, analysis, redesign, implementation and monitoring phases. We modeled the AS-IS and TO-BE states, using BPMN2.0 notation. Quantitative analysis involved cycle time measurements, while qualitative analysis included value-added and flow analyses. The study confirmed the transformative potential of BPM in effectively digitalizing a single case study in the University setting.

While not introducing a novel method, this study holds relevance within the literature by applying the structured approach of the BPM lifecycle of Dumas *et al.* (2018) in a real-world case, demonstrating its practical feasibility and capacity to effect positive change when applied to processes of limited extension and complexity. We explicate and support its application with empirical evidence, confirming results from previous literature and extending the approach to the actual digitalization of a real case study.

Notably, this study provides an in-depth exploration of the main phases of the BPM lifecycle, elucidating the used tools. Unlike most of the examined case studies, this paper addresses the implementation of the proposed intervention and analyses its impact on the organization, illustrating how to overcome resistance to change, which is typical of this type of project. This study underscores the importance of integrating all ISs involved in the process, as this approach streamlines the process, enhancing coordination and improving overall efficiency.

6.1 Implications

This study bridges the gap between theory and practice by demonstrating the practical feasibility of BPM as a methodological approach to digitalizing organizational processes. The successful implementation of BPM in the BTRA process can serve as a model for similar initiatives, in both scope and complexity, in other public and private sector organizations, leading to improved service delivery and increased transparency. Hence, the

study is valuable in comparable settings. The examination of the BPM lifecycle phases, together with the tools used, offers practical guidance for practitioners seeking to replicate it. It highlights the strategies used to manage change due to their inherent cultural resistance.

6.2 *Limitations of the study*

The methodology chosen to address the research hypotheses is the case study, affected by the specific organization and process under examination. It has limitations, such as challenges in generalizing the findings across multiple fields. The main limitation resides in the characteristics of the BTRA process examined, which is less complex and structured as compared to typical public sector processes. The generalizability and broader application of the approach outside of the university sector need further studies and deeper analysis.

6.3 *Recommendations for future research*

The inherently subjective nature of the results (Helfert, 2009), due to the case study methodology, prompts a call for the implementation of a multiple case studies methodology. This approach is a commonly adopted strategy in research (Benbasat *et al.*, 1987) and would facilitate a more comprehensive understanding by exploring varied contexts and extending the research scope to other bodies within the public sector. This intervention can also be extended to more than one organization at a time, enhancing interoperability among organizations.

Future research should analyze more complex processes to verify the extendibility of this approach to public sector processes in general, as they are inherently more complex and structured while this approach is relatively simple.

Notes

1. https://commission.europa.eu/funding-tenders/find-funding/eu-funding-programmes/digital-europe-programme_en
2. www.smartnation.gov.sg
3. www.dta.gov.au
4. <https://hstemberome.treasury.gov/data/digital-government-strategy-milestones>

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Appendix 1. AS-is model

Table A1. Value-added analysis and activities' durations of the AS-IS model

Activity	AS-IS process	Time [min]	Value-added analysis
<i>1 Fill out the authorization form</i>			
1.1 Download trip authorization form		1	NVA
1.2 Fill out form		5	VA
1.3 Download rental request form		1	NVA
1.4 Fill out form		5	VA
1.5 Download request form for own vehicle use		1	NVA
1.6 Fill out form		5	VA
1.7 Download request for advance payment		1	NVA
1.8 Fill out form		10	VA
1.9 Attach estimates		13	VA
2 Send form to trip office		4	NVA
3 Archive document		2	NVA
<i>4 Log document</i>			
5 Send to trip office		301	NVA
<i>6 Finalize first authorization steps</i>			
6.1 Forward documentation to scientific director		3	NVA
6.2 Evaluate		305	BVA
6.3 Affix signature		0,5	BVA
6.4 Forward documentation to trip office		5	NVA
6.5 Send request to PhDs coordinator		3	NVA
6.6 Evaluate		305	BVA
6.7 Affix signature		0,5	BVA
6.8 Forward documentation to trip office		5	NVA
6.9 Send request to fund's responsible		3	NVA
6.10 Evaluate		305	BVA
6.11 Choose fund		2	VA
6.12 Affix signature		0,5	BVA
6.13 Send request to head of fund's affiliation unit		3	NVA
6.14 Evaluate		305	BVA
6.15 Affix signature		0,5	BVA
6.16 Forward of documentation to trip office		5	NVA
7 Send request to dean		3	NVA
8 Send request to general manager		3	NVA
9 Send request to head of department		3	NVA
10 Evaluate		305	VA
11 Insert signature		0,5	VA
12 Forward of documentation to trip office		5	NVA
13 Send documentation to protocol office		5	NVA
<i>14 Log document</i>		2	BVA
15 Send to trip office		301	NVA
<i>16 Manage expense advance request</i>			
16.1 Evaluate expenses' refundability			
16.2 Enter data			
16.2.1 Enter data relating to advance of trip's expenses		14	BVA
16.2.2 Give payment authorization		5	VA
16.3 Issue payment order		10	VA

(continued)

Table A1. Continued

Activity	AS-IS process	Time [min]	Value-added analysis
<i>17 Request trip cancellation</i>			
17.1 Download trip cancellation form		1	NVA
17.2 Fill out trip cancellation form		4	VA
17.3 Send form		3	NVA
17.4 Send to protocol office		2	NVA
<i>17.5 Log document</i>			
17.6 Send request to scientific director		3	NVA
17.7 Evaluate		305	BVA
17.8 Affix signature		0,5	VA
17.9 Forward documentation to trip office		5	NVA
17.10 Send request to fund's responsible		3	NVA
17.11 Evaluate		305	BVA
17.12 Affix signature		0,5	VA
17.13 Forward documentation to trip office		5	NVA
17.14 Send request to head of applicant's affiliation unit		3	NVA
17.15 Evaluate		305	BVA
17.16 Affix signature		0,5	VA
17.17 Forward of documentation to trip office		5	NVA
17.18 Log document			
18 Release sum		10	NVA
19 Send advance refund document		14	VA
20 Download expenses reimbursement form		1	NVA
21 Fill out expenses' reimbursement form		14	VA
22 Send form		4	NVA
23 Archive document		2	NVA
<i>24 Log document</i>			
25 Send to trip office		301	NVA
<i>26 Manage expenses reimbursement request</i>			
26.1 Evaluate expenses' refundability			
26.2 Enter data			
26.2.1 Enter data relating to trip		20	BVA
26.2.2 Enter expenses (description + amount)		14	BVA
26.2.3 Enter advance report		3	BVA
26.2.4 Give payment authorization		5	VA
26.3 Issue payment order		10	VA
27 Send notification to applicant of inability to make reimbursement		3	NVA

Source: Authors' own work

Table A2. Value-added analysis and activities' durations of log document sub-process

Activity	Log document		Value-added analysis
	Time [min]		
Upload document(s) to document management system	2		NVA
Log document	0		BVA
Source: Authors' own work			

Table A3. Value-added analysis and activities' durations of evaluate expenses' refundability sub-process

Activity	Evaluate expenses' refundability		Value-added analysis
	Time [min]		
Evaluate expenses' refundability	20		VA
Notify applicant of non-reimbursable expenses	3		NVA
Send request	3		NVA
Evaluate	305		BVA
Affix signature	0,5		BVA
Send request	3		NVA
Evaluate	305		BVA
Affix signature	0,5		BVA
Send request	3		NVA
Source: Authors' own work			

Appendix 2. TO-BE model

Table A4. Activities' durations of the TO-BE model

Activity	TO-BE process	Time [min]
<i>1 Authenticate</i>		
2 Log in to trip area		0
<i>3 Request trip authorization</i>		
3.1 Enter type, location, estimated cost, start and end date of trip		2
3.2 Select fund's responsible		1
3.3 Check on "IT'S ME"		0,1
3.4 Select fund		1,5
3.5 Select scientific director		1
3.6 Select PhDs coordinator		1
<i>3.7 Request to use your own vehicle</i>		
3.7.1 Enter route data		0,5
3.7.2 Declare reason for own vehicle's use		1
3.7.3 Enter car and driving license data		1,5
3.7.4 Select car		0,5
<i>3.8 Request rental</i>		
3.8.1 Select car type		0,5
3.8.2 Enter motivation		1
<i>3.9 Request advance expenses</i>		
3.9.1 Notify impossibility to complete request		0
3.9.2 Enter accommodation estimate		0,5
3.9.3 Attach quote		2
3.9.4 Enter travel estimate		0,5
3.9.5 Attach quote		2
3.9.6 Enter fee		0,5
3.9.7 Attach quote		2
3.9.8 Send request		0,1
3.10 Generate report		0
3.11 Send request		0,1
<i>3.12 Evaluate</i>		
3.13 Affix signature		0,2
<i>3.14 Evaluate</i>		
3.15 Affix signature		0,2
3.16 Evaluate		
3.17 Select fund		2
3.18 Affix signature		0,2
<i>3.19 Evaluate</i>		
3.20 Affix signature		0,5
<i>3.21 Evaluate</i>		
3.22 Affix signature		0,2
3.23 Log document		0
4 Send request to dean		0
5 Send request to general manager		0
6 Send request to head of department		0
(continued)		

Table A4. Continued

TO-BE process	
Activity	Time [min]
<i>7 Final authorization</i>	
7.1 Authenticate	
7.2 Log in to trip area	0
7.3 Select trip	0,5
7.4 Evaluate	185
8 Insert digital signature	0,5
9 Produce authorization document	0
10 Log document	0
<i>11 Manage request for advance payment of expenses</i>	
11.1 Evaluate expenses' refundability	
<i>11.2 Enter data</i>	
11.2.1 Authenticate	
11.2.2 Enter data of advance of trip's expenses	20
11.2.3 Give payment authorization	5
<i>11.3 Issue payment order</i>	
<i>12 Request trip cancellation</i>	
<i>12.1 Authenticate</i>	
12.2 Log in to trip area	0
12.3 Fill out cancellation request	5
12.4 Send request	0,1
12.5 Log document	0
12.6 Send request to scientific director	0
<i>12.7 Authenticate</i>	
12.8 Log in to trip area	0
12.9 Select trip	0,5
12.10 Evaluate cancellation request	185
12.11 Affix signature	0,2
12.12 Send request to fund's responsible	0
<i>12.13 Authenticate</i>	
12.14 Log in to trip area	0
12.15 Select trip	0,5
12.16 Evaluate cancellation request	185
12.17 Affix signature	0,2
12.18 Send request to administrative manager	0
<i>12.19 Authenticate</i>	
12.20 Log in to trip area	0
12.21 Select trip	0,5
12.22 Evaluate cancellation request	185
12.23 Affix signature	0,2
12.24 Send request to head of applicant's affiliation unit	0
<i>12.25 Authenticate</i>	
12.26 Log in to trip area	0
12.27 Select trip	0,5
12.28 Evaluate cancellation request	185
12.29 Affix signature	0,2
12.30 Log document	0
(continued)	

Table A4. Continued

Activity	TO-BE process	Time [min]
<i>13 Authenticate</i>		
14 Recover advance		14
15 Release sum		0
<i>16 Request reimbursement of expenses</i>		
16.1 Authenticate		
16.2 Log in to trip area		0
16.3 Open refund request section		0,2
16.4 Choose between flat-rate and analytical reimbursement		0,5
<i>16.5 Enter items and invoices</i>		
16.5.1 Choose expense's type		0,5
16.5.2 Choose car		0,5
16.5.3 Enter itinerary		0,7
16.5.4 Enter amount		0,5
16.5.5 Enter expense attachment		2
16.5.6 Enter departure and arrival dates		0,5
16.5.7 Enter departure and arrival dates		0,5
16.5.8 Enter amount		0,5
16.5.9 Enter expense attachment		2
16.5.10 Enter amount		0,5
16.5.11 Enter expense attachment		2
16.6 Prepare enter key		0
16.7 Send request		0,1
17 Enter start and end dates of trip and reasons		1
18 Log document		0
<i>19 Manage expense reimbursement request</i>		
19.1 Evaluate expenses' refundability		
19.2 Enter data		
19.2.1 Authenticate		
19.2.2 Enter data relating to trip		14
19.2.3 Enter expenses (description + amount)		14
19.2.4 Enter advance report		3
19.2.5 Give payment authorization		5
19.3 Issue payment order		10

Source: Authors' own work**Table A5.** Activities' durations of authenticate sub-process

Activity	Authenticate	Time [min]
Enter ID and password		0,5
Check credentials		0
Authorize to access and identify user data		0
Deny access		0

Source: Authors' own work

Table A6. Activities’ durations of evaluate sub-process

Evaluate	
Activity	Time [min]
<i>Authenticate</i>	
Log in to trip area	0
Select trip	0,5
Evaluate	185
Source: Authors’ own work	

Table A7. Activities’ durations of evaluate expenses’ refundability sub-process

Evaluate expenses’ refundability	
Activity	Time [min]
<i>Authenticate</i>	
Evaluate expenses’ refundability	0,5
Notify applicant of non-reimbursable expenses	20
	3
<i>Authenticate</i>	
Log in to trip area	0
Select trip	0,5
Evaluate	185
Affix signature	0,2
<i>Authenticate</i>	
Log in to trip area	0
Select trip	0,5
Evaluate	185
Affix signature	0,2
Source: Authors’ own work	

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